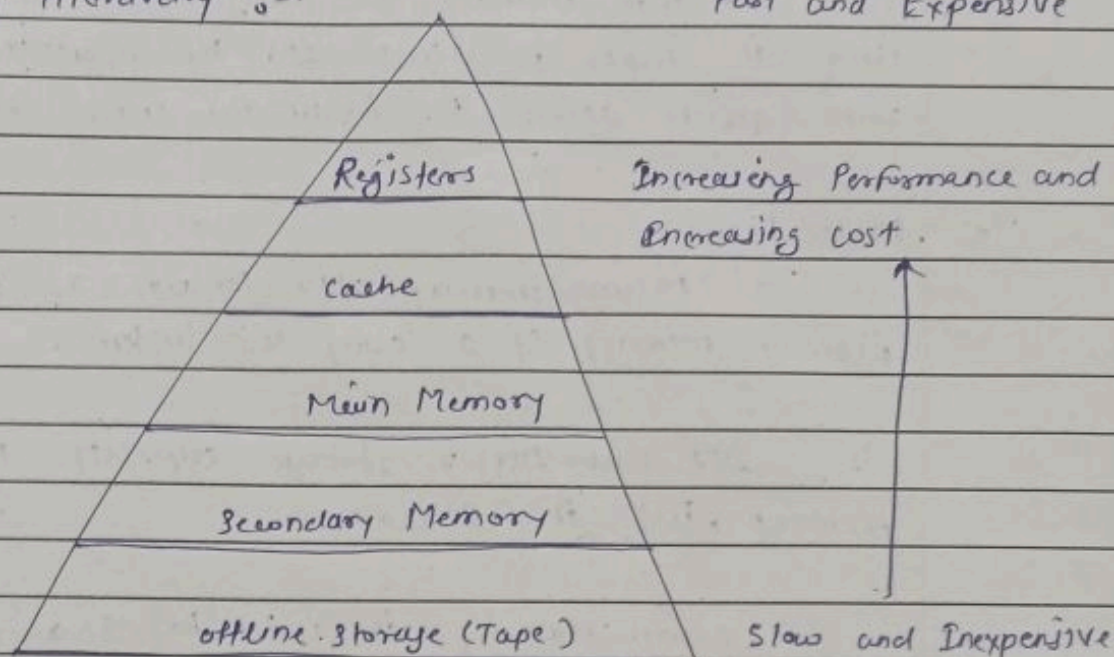


Memory Organization

* Memory Hierarchy :-



1. Registers :-

- Registers are small, high speed memory units located in the CPU.

- They are used to store the most frequently used data and instructions.

- Registers have the fastest access time and the smallest storage capacity, typically ranging from 16 to 64 bits.

2. Cache Memory :-

- Cache memory is a small, fast memory unit located close to the CPU.

- It stores frequently used data and instructions that have been recently accessed from the main memory.

- Cache memory is designed to minimize the time it takes to access data by providing the CPU with quick access to frequently used data.

3. Main Memory :-

- Main memory also known as RAM, is the primary memory of a computer system.

- It has larger storage capacity than cache memory, but it is slower.

- Main memory is used to store the data and instructions that are currently used ~~to store data~~ by CPU.

4. Secondary Memory :-

- Secondary memory such as HDD, Solid-State drives (SSD) is a non-volatile memory unit that has a larger storage capacity than main memory.

- It is used to store data and instructions that are not currently in use by CPU.

- Secondary storage has the slowest access time and is typically the least expensive type of memory in memory hierarchy.

5. Magnetic Disk :- (Magnetic Tape)

- It is generally used for the backup of data.

- In the case of magnetic tape, the access time for a computer is little slower and therefore it requires some amount of time for accessing the strip.

* Characteristics of Memory Hierarchy

1. Capacity
2. Access time
3. Performance
4. Cost per Bit

* Advantages :

- It helps in removing some destruction, and managing the memory in a better way.

- It helps in spreading the data all over the computer system.

- It saves the consumer's price and time.

Memory chip organization :-

Memory is divided into cells and each of them is identified by a unique number called an address.

When CPU wants to read or write an address, it generates control signals such as "read" and "write", which each cell can identify.

Memory Request :-

- Memory request contains the address along with the control signals.

- When inserting data into the stack, each block consumes memory and the number of memory cells can be determined by the capacity of a memory chip.

Q. Find total no. of cells in $64k \times 8$ memory chip.

eg. size of each cell = 8

$$\text{No. of bytes in } 64k = 2^6 \times 2^{10}$$

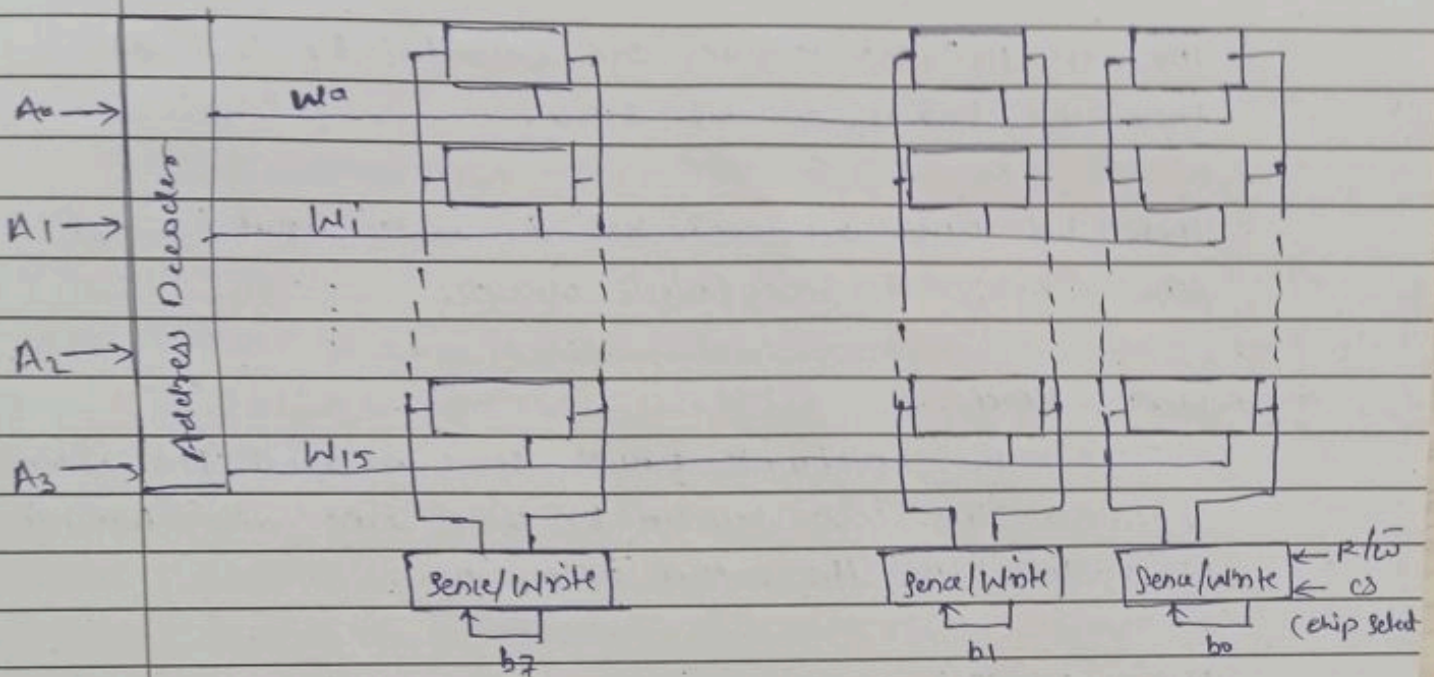
$$\therefore \text{Total Number of cells} = 2^{16} \text{ cells.}$$

Word Size :-

- It is the maximum number of bits a CPU can process at a time and it depends upon the processor.

- Word size is fixed size piece of data handled as a unit by the instruction set or the hardware of a processor.

16x8 Memory Organization:-



16 Memory locations: $W_0, W_1, \dots, W_{15}$

8 bits in each location: $b_0, b_1, \dots, b_7$

- A memory cell is capable of storing 1-bit of information.
- A number of memory cells are organized in the form of a matrix to form the memory chip.
- Each row of cells contains a memory word.
- All cells of row are connected to a common line which is referred as word line.
- An address decoder is used to drive the word line.
- At particular instance, one word line is enabled depending on the address present in the address bus.

- The cell in each column are connected by two lines. These are known as bit lines.

- These bit lines are connected to data input/output lines through a sense/write circuit.

- During Read :-

the sense/write circuit sense or read the information stored in the cells selected by word line and transmit this info. to the output data line.

- During Write :-

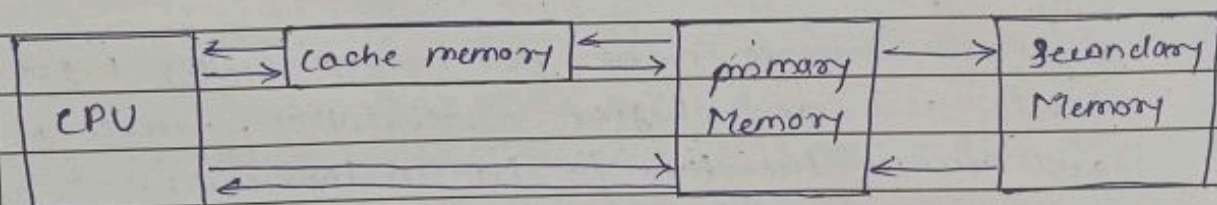
the sense/write circuit receives the information and store it in the cells of the selected word.

eg. - for 16 words we need an address bus of size 4. (2^4)

- The CS (chip select) line is required to select a given chip in multi chip memory system.

* Cache Memory :-

- Cache memory is an extreme fast memory type that acts as buffer between RAM and CPU.
- Cache memory holds frequently requested data and instructions so that they are immediately available to the CPU when needed.
- Cache memory is costlier than main memory or disk memory but more economical than CPU registers.
- cache memory is used to speed up and synchronize with a high-speed CPU.



- Cache Performance :-

when processor needs to read or write a location in main memory, it first checks for a corresponding entry in cache.

- ① if the processor finds the memory location that is in cache (present in cache), a cache Hit has occurred and data is read from the cache.
- ② If the processor does not find the memory location in the cache, a cache miss has occurred.

for a cache miss, the cache allocates a new entry and copies in data from the main memory, then the request is fulfilled from the contents of cache.

The performance of cache memory is frequently measured in terms of a quantity called Hit Ratio.

$$\text{Hit Ratio} = \frac{\text{Hit}}{(\text{Hit} + \text{miss})} = \frac{\text{No. of Hits}}{\text{total accesses}}$$

$$\text{Miss Ratio} = \frac{\text{miss}}{(\text{Hit} + \text{miss})} = \frac{\text{No. of miss}}{\text{total accesses}} = 1 - \text{Hit Ratio} \quad (H)$$

we can improve cache performance using higher cache block size and higher associativity, reduce miss rate, and reduce the time to hit in the cache.

• Cache Mapping

There are three different types of mapping used.

- ① Direct Mapping
- ② Associative Mapping
- ③ Set-Associative Mapping.

Cache Read

data is in the cache



Forward to CPU

Data is not in the cache



Load Through: forward the word as cache line is filled

or

Fill cache line and then forward word.

Cache Write

Data is in the cache



Write Through:

Write data to both cache and main memory

or

Write Back: Write data to cache only.

Defer main memory write until block is flushed.

Data is not in the cache



Write Allocate: Bring line into cache, then update it.

or

Write No-Allocate: update main memory only.

* Virtual Memory :-

- Virtual memory is a common technique used in a computer OS.
- Virtual memory is a method that computers use to manage storage space to keep systems running quickly and efficiently.
- It creates an illusion that the system has more memory than it actually does, by dividing the memory into fixed-size units called pages and storing some of them on disk when they are not needed.
- virtual memory free up RAM by swapping data that has not been used recently over to a storage device, such as hard drive or solid-state drive (SSD).
- virtual memory is important for improving system performance, multitasking and using large programs.
- Advantages :-
 - More processes may be maintained in the main memory.
 - A process may be larger than all of the main memory can run easily.
 - Allows greater multiprogramming levels by using less of available memory.
 - It makes it possible for several larger applications to run at once.

- Disadvantages :-

- It can slow down the system performance, as data needs to be constantly transferred between the physical memory and the hard disk.
- It can increase the risk of data loss or corruption, as data can be lost if the hard disk fails.
- It can increase the complexity of memory management system, as OS needs to manage both physical and virtual memory.

* Paging :-

Virtual memory breaks program into fixed-size blocks called pages. The pages are normally an integral power of two in size such as $2^{10} = 1024$ bytes.

Page #	Present Bit	Disk Address	Page frame
0	1	10110111011	00
1	0	10011000011	XX
2	1	11000111011	01
3	0	11101111011	XX
4	1	01110111011	11
5	0	10111110111	XX
6	0	10101010111	XX
7	1	11011011011	10

Present Bit :-

- 0: Page is not in physical memory
- 1: Page is in physical memory.

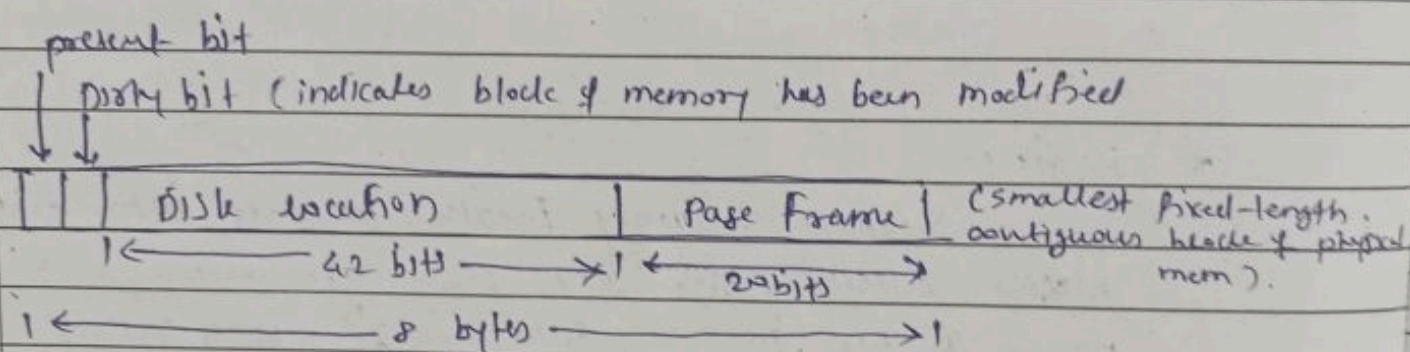
* Page fault:

If required page is not present in physical memory then page fault is occurred.

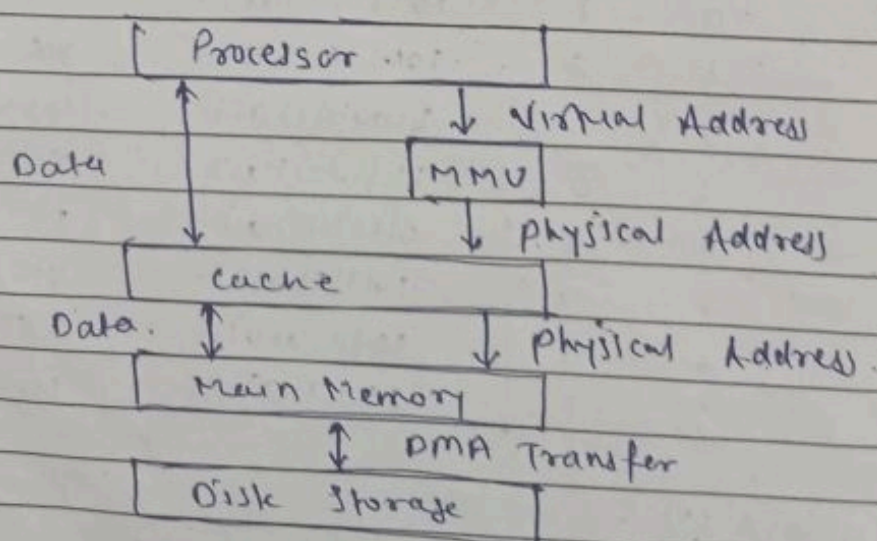
- when page fault occurs CPU will fetch page into physical memory using its virtual address.

- The demand paging scheme does not load a page into memory until there is a page fault.

A page table entry occupy 8 bytes as:



The working of virtual memory is as belows:



DMA: direct memory Access
mmu: Memory Management Unit

* Parallel Bus Architecture :-

- A computer system may contain many components that need to communicate with each other. In a worst-case all N components may need to simultaneously communicate with every other component, in which $N \times (N-1)/2$ links are needed for N components.

- A bus is a common pathway that connects a number of devices. (eg. mother-board of a personal computer).

• Bus Structure, Protocol and control :

A bus consists of physical parts like connectors and wires and a bus protocol.

The wires can be partitioned into separate groups of control, address, data and power.

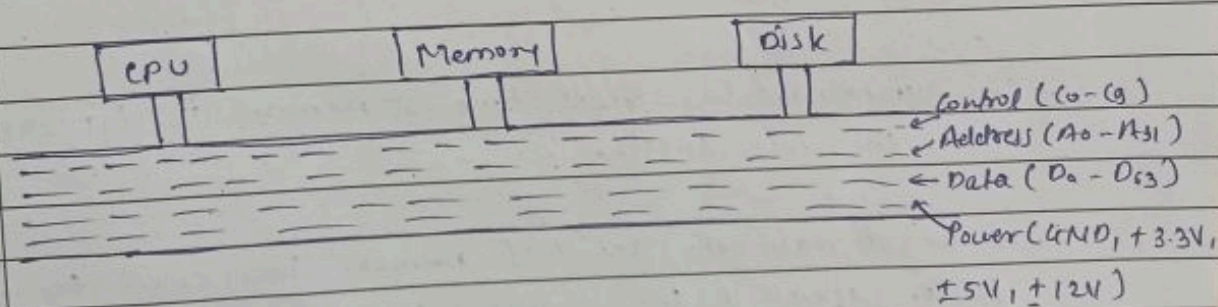


Fig. Simplified illustration of Bus.

The devices share a common set of wires, and only one device may send data at any one time. All devices simultaneously listen, but normally only one device receives.

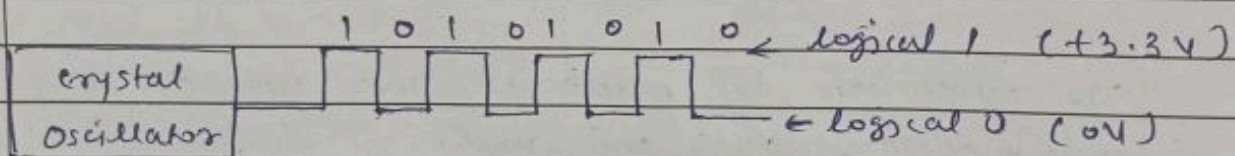
- only one device can be bus master and remaining devices are then considered to be slaves.
- The master controls the bus and can be either sender or receiver.

A Bus can be classified as one of two types:

1. Synchronous
2. asynchronous

- For synchronous bus, one of the device that is connected to bus contains an oscillator (a clock) that sends out a sequence of 1's and 0's at timed intervals.

Bus clocking :-



- In synchronous bus read time of memory is typically slower than the bus speed.
- if we increase the speed of ~~memory~~ bus clock to match faster speed of the memory, then slower devices that uses the bus clock may not work properly.
- An asynchronous bus solves this problem, but is more complex, because there is no bus clock.

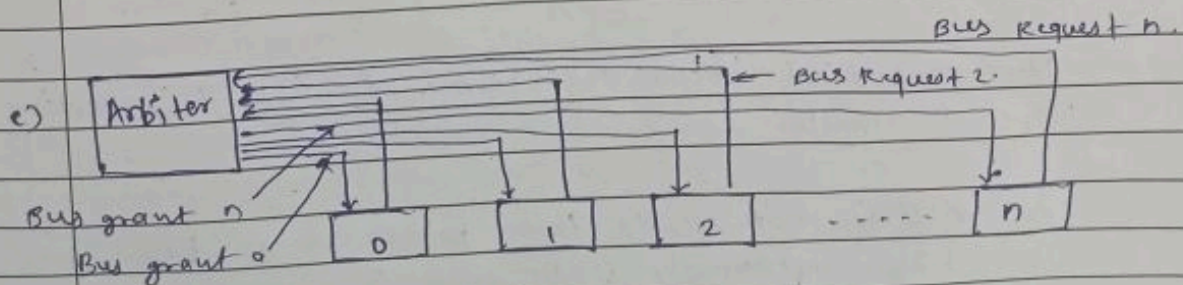
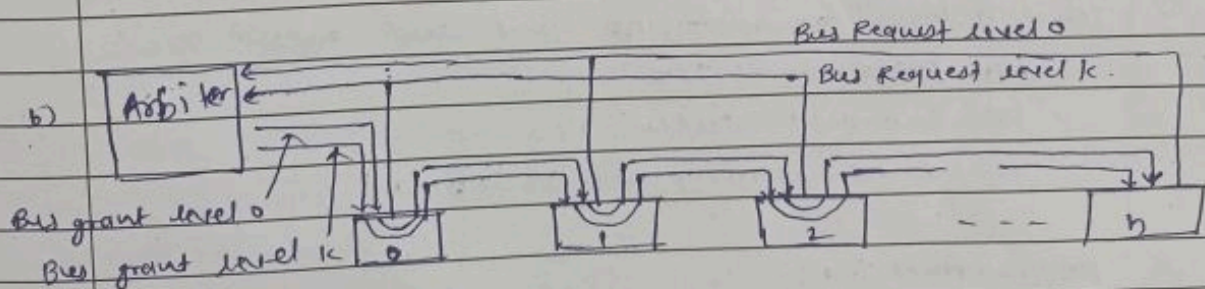
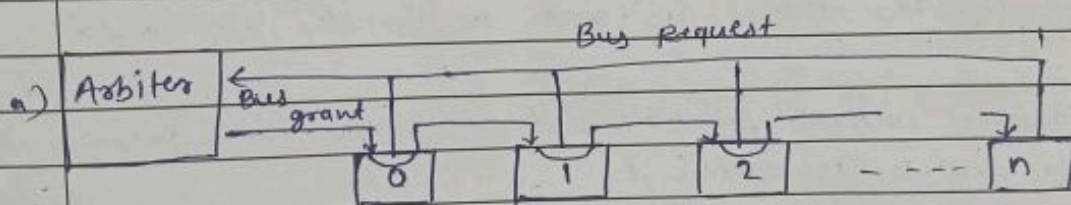
- Asynchronous buses can be more difficult to debug than synchronous buses when there is a problem and interfaces for asynchronous buses can be more difficult to make.

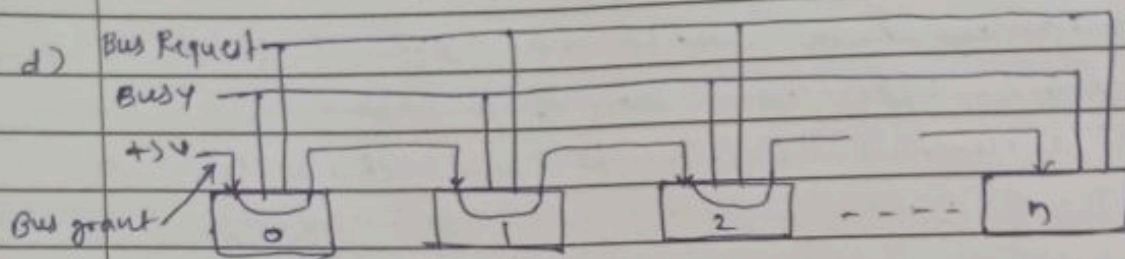
For these reasons, synchronous buses are very common, particularly in personal computers.

• Bus Arbitration - Master and Slaves :-

suppose, more than one device wants to be a bus master at the same time. How is a decision made as to who will be the bus master? this is bus arbitration problem.

there are two basic schemes: centralized and decentralized.





- fig
- Simple centralized Bus Arbitration
 - Centralized Arbitration with priority levels.
 - Fully centralized bus arbitration
 - decentralized bus arbitration

* Internal Communication Methodologies :-

cpu and ~~memory~~ computer system communicates with the memory and the I/O devices in order to transfer the data.

- However method of communication of cpu with memory and I/O devices is different.

- Three methods for managing input and output are :

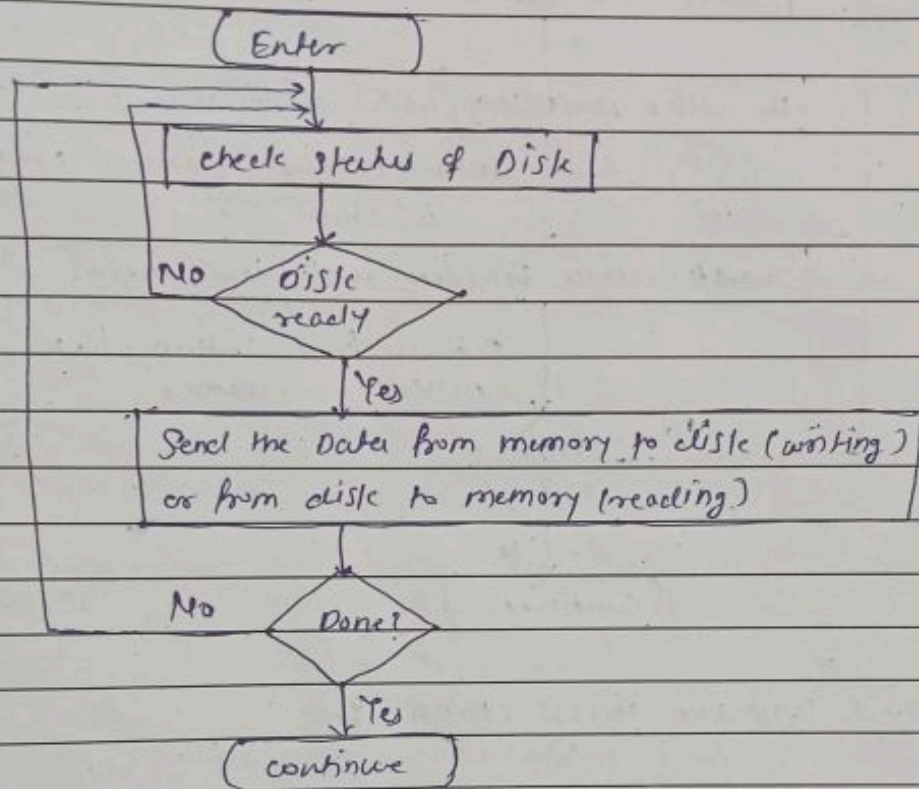
- ① programmed I/O
- ② Interrupt-driven I/O
- ③ direct memory access (DMA)

① programmed I/O :-

- The cpu first checks the status of disk by reading special register.

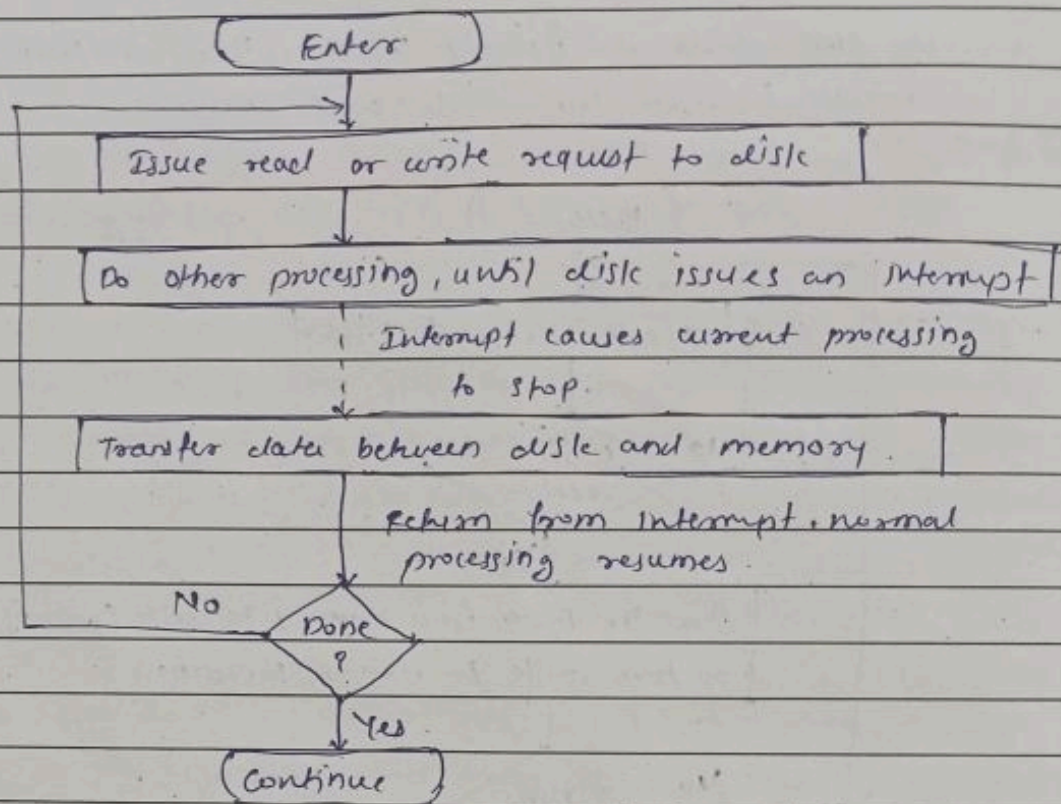
- if disk is not ready then process loops back and checks the status continuously until disk is ready. this is referred as "busy-wait."

- when the disk is finally ready, then a transfer of data is made between the disk and CPU.



② Interrupt-Driven I/O :-

- The CPU issues a request to the disk for reading or for writing, and then immediately resumes execution for another process.
- At some time, when the disk is ready it interrupts the CPU. the CPU then invokes an Interrupt service routine (ISR) for disk and return to normal execution when (ISR) completes its task.
- Then transfer data between disk and memory.



③ Direct Memory Access (DMA) :-

- A direct memory access (DMA) device can transfer data directly to and from memory, rather than using CPU as an intermediary, and can thus relieve congestion on the system bus.

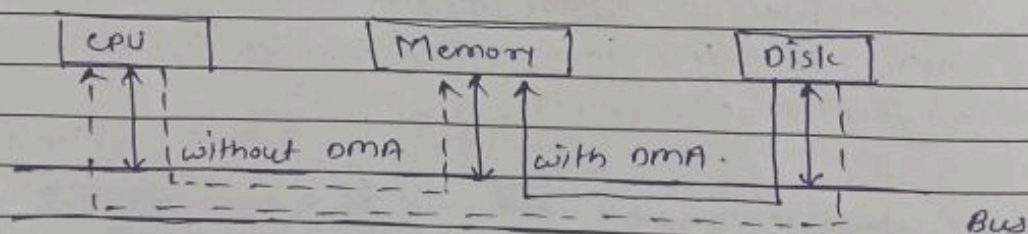
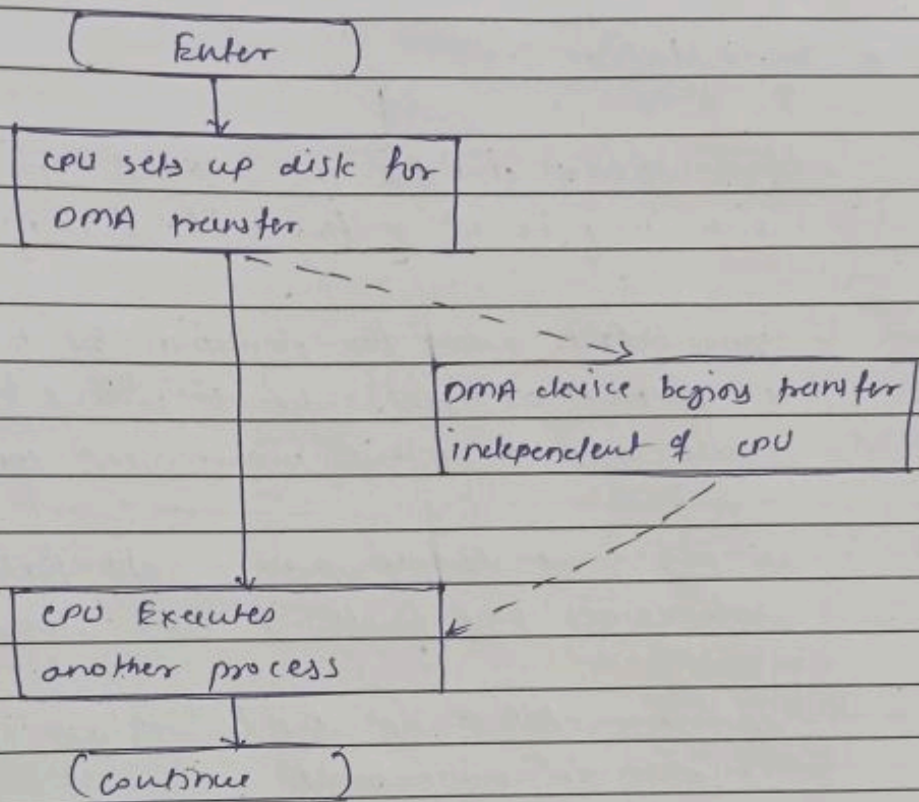


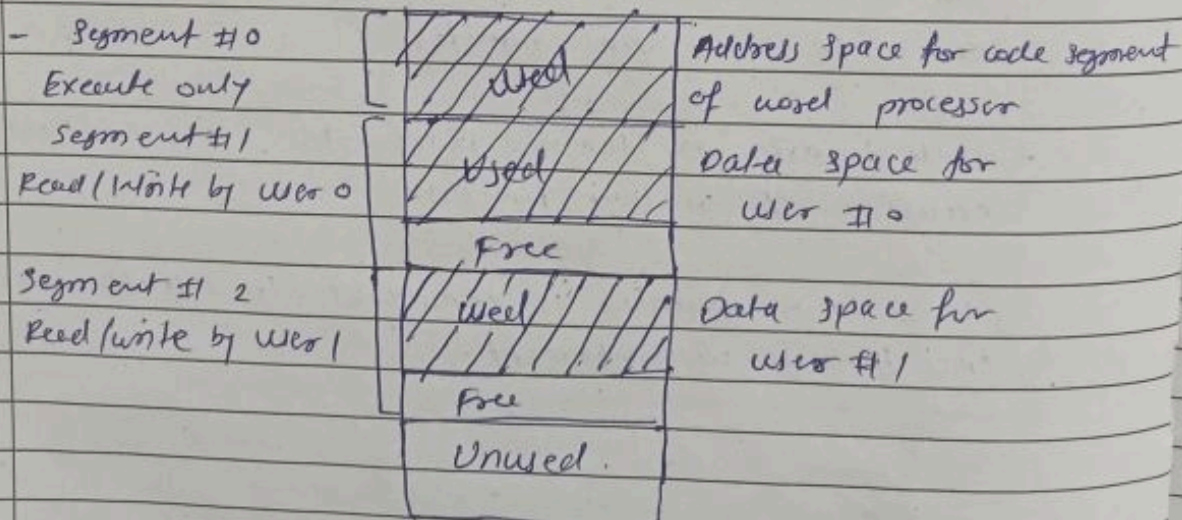
Fig. DMA transfer from disk to memory bypass the CPU.



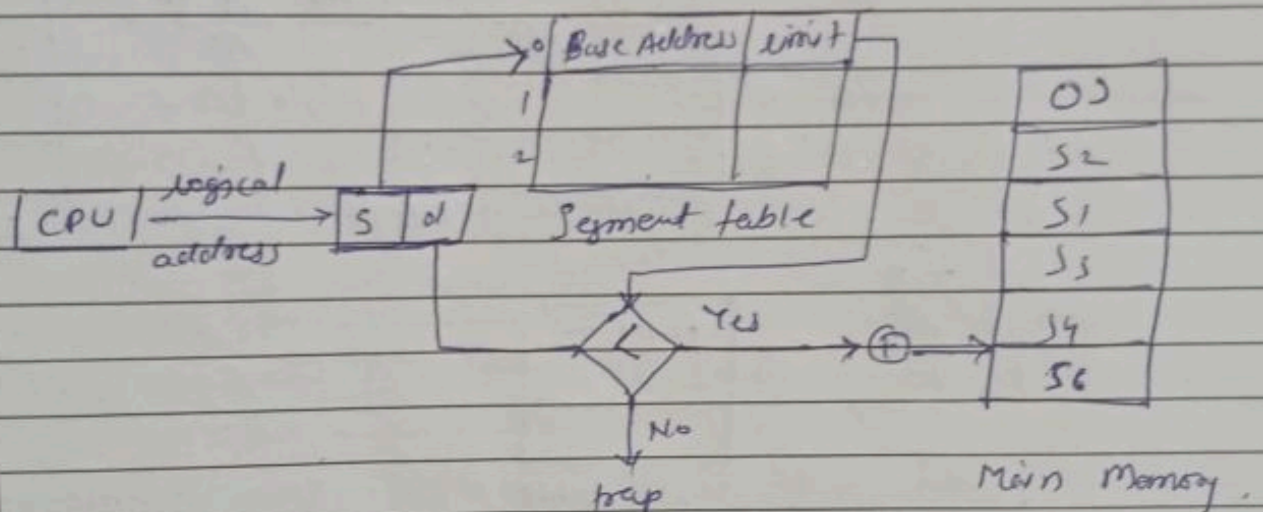
- The CPU sets up the DMA device and then signals the device to start the transfer.
- while transfer is taking place, the CPU continues execution of another process.
- when DMA transfer is completed, the device informs the CPU through an interrupt.

* Segmentation :-

- Segmentation divides the address space into segment, which may be of different / (Arbitrary) size.
- Segmentation allow for protection, so that segments may be specified as 'read only' to prevent changes or execute only to prevent unauthorized copying
- In paging we divide memory into fixed size pages which causes page fault or error while accessing data.
- In segmentation we divide address space as per the user requirement.
- Segment containing Base address and size of segment.



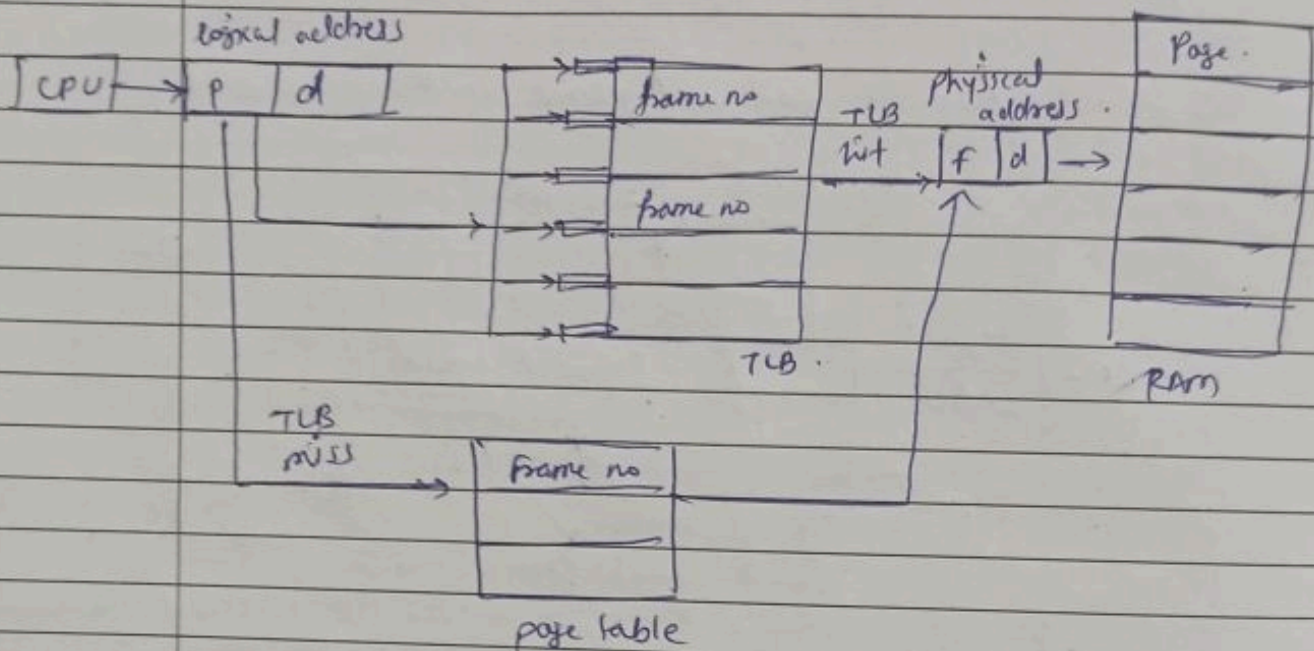
- In order to specify an address in segmented memory, the users program must specifying a segment number and address within the segment. The operating system then translates the user's segmented address to a physical address.
- segmentation is non-contiguous memory allocation.



S - segment No / ~~offset~~ (~~Base address~~)
 d - offset / size

* Translation lookaside Buffer :-

Indirectly ~~the~~ TLB, we call it as cache.



The TLB holds 8 entries for a system that has 32 pages and 16 page frames. The virtual page field is 5 bits wide because $2^5 = 32$.

likewise physical page field is 4 bits wide because $2^4 = 16$.

Valid	Virtual Page Number	Physical Page Number
1	01001	1100
1	10011	1010
0	11011	1001
0	10101	0110
1	01101	1000

* Page Replacement Algorithm :-

In paging processes are divided into pages and main memory is divided into frames.

- Pages of process are loaded into frames of main memory when required.
- Page Replacement algorithm is used when page fault occurs, i.e. page referenced by CPU is not present in the main memory.
- Page Replacement algorithm is used to decide which page will be replaced to allocate memory to the referenced page.
- Different page replacement algorithms suggest different way to decide which page is to be replaced.
- The main objective of these algorithms is to reduce the number of page faults.

① First In First Out (FIFO) :-

- This algorithm is similar to the operations of the queue.
- All the pages are stored in the queue in the order they are allocated in frames in the main memory.
- The one which is allocated in the memory first is replaced first.

- The one which is at the front of the queue is removed at the time of replacement.

example: consider the page referenced by the CPU in the order are 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, 7, 9, 6

Pages $\Rightarrow$	6	7	8	9	6	7	1	6	7	8	9	1	7	9	6
Frame 3			8	8	8	7	7	7	7	7	9	9	9	9	9
Frame 2		7	7	7	6	6	6	6	6	8	8	8	8	8	6
Frame 1	6	6	6	9	9	9	1	1	1	1	1	1	7	7	7
	miss	miss	miss	miss	miss	miss	miss	hit	hit	miss	miss	hit	miss	hit	miss

2. Least Recently Used :-

- LRU is based on strategy that whenever a page fault occurs, the least recently used page will be replaced with a new page.

- So the page not utilized for the longer time in the memory gets replaced.

example: consider Pages referenced by CPU in order 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, 7, 9, 6.

pages $\Rightarrow$	6	7	8	9	6	7	1	6	7	8	9	1	7	9	6
Frame 3			8	8	8	7	7	7	7	7	7	1	1	1	6
Frame 2		7	7	7	6	6	6	6	6	6	9	9	9	9	9
Frame 1	6	6	6	9	9	9	1	1	1	8	8	8	7	7	7
	miss	miss	miss	miss	miss	miss	miss	hit	hit	miss	miss	miss	hit	hit	miss

3) Most Recently Used (MRU) :-

- In this algorithm, page will be replaced which has been used recently.

- Belady's anomaly can occur in this algorithm.

example:- Consider page referenced by CPU in order 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, 7, 9, 6, 7, 8, 9, 1, 7, 9, 6

pages →	6	7	8	9	6	7	1	6	7	8	9	1	7	9	6
frame 3			8	9	9	9	9	9	9	9	9	9	9	9	6
2		7	7	7	7	7	1	1	1	1	1	1	7	7	7
1	6	6	6	6	6	6	6	6	7	8	8	8	8	8	8
	miss	miss	miss	miss	hit	hit	miss	hit	miss	miss	hit	hit	miss	hit	miss

4) Optimal Page Replacement (OPR) :-

- OPR algorithm replaces the page which is not used in longest dimension of time in future.

example:- Consider page referenced by CPU in order 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, 7, 9, 6, 7, 8, 9, 1, 7, 9, 6

pages	6	7	8	9	6	7	1	6	7	8	9	1	7	9	6
frame 3			8	9	9	9	1	1	1	1	1	1	1	1	6
2		7	7	7	7	7	7	7	7	7	7	7	7	7	7
1	6	6	6	6	6	6	6	6	8	9	9	9	9	9	9
	miss	miss	miss	miss	hit	hit	miss	hit	hit	miss	miss	hit	hit	hit	miss